NexusMart - Data Quality Report

CODE//CONNECT'26 Data Analytics Module · Phase 1 submission · generated automatically by `scripts/build_dq_report.py` from the pipeline's own logs (rule log, reject table, quality checks).

863,000
raw rows across 9 files

93
cleaning rules that fired

361,566
values standardised, repaired, imputed or merged

31/31
automated quality checks passed

1. Principles

- **All cleaning is code** (`nexusmart_etl/clean.py`); nothing was edited by hand, in Excel, in the database or in Power Query.
- **Nothing is silently deleted.** Every row that leaves a table is written to the reject table with a reason code. Value repairs are flagged on the row (`dq_flags` column in FactSales, `*_imputed` / `*_repaired` columns in staging).
- **Repair from evidence first** (other columns or tables of the same record), impute second, quarantine only if neither is possible.
- **Thresholds are documented in code** and in this report; the Airflow DAG fails when a critical quality threshold is breached.

2. Before and after

Table	Raw rows	Exact duplicates removed	Duplicate entities merged	Other quarantined	Clean rows	Values repaired / imputed*
locations	650	0	0	0	**650**	50
sellers	420	0	0	0	**420**	22
products	3,200	0	25	0	**3,175**	1,451
customers	45,000	0	2,500	0	**42,500**	4,667
orders	163,200	3,200	0	0	**160,000**	7,687
order_items	315,143	0	0	0	**315,143**	36,528
payments	168,111	3,296	0	0	**164,815**	663
shipping	151,228	0	0	0	**151,228**	9,636
returns	16,048	0	0	15	**16,033**	2,855

* Count of value-level fixes (one row can receive several). Every removed row is stored in `dq.reject_records` / `data/rejects/reject_records.csv` with a reason code.

3. What the raw data looked like

File	Rows	Exact dup rows	Dup IDs	Blank values by column
customers	45,000	0	0	birth_year 4,600 (10.2%), phone 3,599 (8.0%), city_id 2,233 (5.0%), gender 2,199 (4.9%), email 1,698 (3.8%)
products	3,200	0	0	sub_category 485 (15.2%), brand 392 (12.2%), weight_kg 296 (9.2%), cost_price 170 (5.3%)
sellers	420	0	0	join_date 34 (8.1%), rating 33 (7.9%), seller_tier 22 (5.2%)
locations	650	0	0	latitude 33 (5.1%), longitude 31 (4.8%), province 25 (3.8%)
orders	163,200	3,200	3,200	discount_code 112,059 (68.7%), ship_city_id 6,516 (4.0%)
order_items	315,143	0	0	discount_amount 31,567 (10.0%)
payments	168,111	3,296	3,296	payment_date 5,017 (3.0%)
shipping	151,228	0	0	delivered_date 18,841 (12.5%), shipping_fee 9,145 (6.0%), shipped_date 5,943 (3.9%)
returns	16,048	0	0	return_reason 2,390 (14.9%)

Main problem types found: 3 date formats mixed in every date column (ISO, DD/MM/YYYY, "Mon D YYYY") plus impossible dates (2024-02-30, month 13, 00/00/2024) and order dates typed as 2026/2027; 74 spellings of 12 categories; 20 spellings of 5 order statuses; 22 payment-method, 22 courier and 38 return-reason variants; amounts stored as text ("Rs. 2,290"); sentinel 9,999,999 in prices, totals and fees; prices recorded 100x too high or in US dollars (~1/280 of the PKR value); negative quantities/prices; quantities written as words or 500; blank discount amounts; discounts larger than the line value; refunds 1.4x the item value; duplicated orders and payments; the same customers and products under different IDs; return IDs with lost zero-padding.

4. Key decisions and assumptions

- **Dates.** Slashed dates are day/month/year (brief). Order dates that are impossible or outside 1 Jan 2024 - 30 Jun 2025 (1,605 orders) were repaired from the earliest payment or shipment date of the same order (validated on clean orders: same month in 97%

of cases, median error 0 days). 0 orders could not be repaired.

- **Unit price.** Selling prices follow list price x a smooth inflation index (+0.25% per month, fitted on clean lines). Prices ~100x the expected value were divided by 100; negative prices made positive; prices in USD, zero, or 9,999,999 were solved exactly from `order_total` when that line was the only unknown in its order (927 lines), otherwise list price x index (16 lines).
- **Quantity.** Words and "2.0" converted; negatives made positive; 0 or >50 (the 500 entry error) solved from `order_total` (627 lines) or set to 1 (14 lines).
- **Discounts.** The order's `discount_code` is the authoritative %. Line discount = % x line gross when blank (31,567), larger than the line (871) or inconsistent (1). After all fixes only 7 orders differ from their `order_total` by more than 1%.
- **Duplicate customers** (2,500 merged, 45,000 → 42,500): same normalised name AND same e-mail or phone, and no conflicting birth year or city. Never on name alone. Orders are re-pointed through `dq.customer_id_map`.
- **Duplicate products** (25 merged): same name and category AND same cost or weight AND list price within 5%. Mapping kept in `dq.product_id_map`.
- **Broken references** are mapped to the *Unknown* member (key -1) instead of being dropped, so revenue is kept: 932 lines with product IDs not in the catalogue, 321 orders with unknown customers. Return IDs with lost zero-padding were re-padded (139).
- **Ship city.** Blank ship city filled with the customer's registered city (6,082 orders, flagged; 89% of orders ship to the registered city). The flag lets the dashboard exclude them.
- **Returns.** Only non-Rejected (incl. "Declined") returns are valid. Refunds above the line's net value are capped at it (321). Returns against cancelled lines are quarantined (15).
- **Shipping.** Missing, negative or 9,999,999 fees were imputed with the median fee of the same courier and order-weight band (9,311 orders). Delivery durations that are negative (1,269) or >30 days (266) are set to NULL, not guessed. One fee per order is allocated to its lines by gross-value share.
- **Products.** Cost above 1.2x list price (an injected 1.5x pattern, 67 products) and blank costs (170) were re-imputed from the sub-category's median cost/list ratio. Invalid list prices were re-derived from the product's own sales.

5. Rule-by-rule log (from `dq.rule_log`)

locations

Column	Rule	What was wrong → what we did	Rows	Action	Before → after
province	STD	Standardised 22 raw variants to 8 canonical values	121	STANDARDISE	
province	STD_BLANK	Blank → 'Unknown'	25	IMPUTE	
latitude/longitude	GEO_RANGE	Coordinates outside Pakistan bounding box → NULL	8	FLAG	
latitude/longitude	GEO_MISSING	Missing coordinates retained as NULL (not used for keys)	69	RETAIN	
province	PROVINCE_KNN	Blank province inferred from 5 nearest cities in same region	25	IMPUTE	

Checked, nothing found: DUP_EXACT, STD, PROVINCE_UNRESOLVED

sellers

Column	Rule	What was wrong → what we did	Rows	Action	Before → after
seller_tier	STD	Standardised 11 raw variants to 4 canonical values	91	STANDARDISE	
seller_tier	STD_BLANK	Blank → 'Unknown'	22	IMPUTE	
join_date	DATE_INVALID	Unparseable/impossible join_date → NULL	4	FLAG	
join_date	DATE_BLANK	Blank join_date retained as NULL	34	RETAIN	
rating	RATING_RANGE	Rating outside 1-5 (e.g. 0, 11) → NULL	22	FLAG	
rating	RATING_BLANK	Blank rating retained as NULL	33	RETAIN	

Checked, nothing found: DUP_EXACT, DATE_RANGE, FK_LOCATION

products

Column	Rule	What was wrong → what we did	Rows	Action	Before → after
category	STD_CATEGORY	74 raw category labels → 12 canonical categories	1,650	STANDARDISE	
sub_category	SUBCAT_BLANK	Blank sub_category → 'Unknown'	485	IMPUTE	
brand	BRAND_FROM_NAME	Blank brand repaired from first word of product_name (known brand)	387	REPAIR	
brand	BRAND_BLANK	Blank brand not recoverable → 'Unknown'	5	IMPUTE	
list_price	PRICE_NEGATIVE	Negative list_price → absolute value (sign error)	4	REPAIR	
list_price	PRICE_INVALID	list_price zero / 9999999 sentinel / non-numeric → re-derived from sales history	9	REPAIR	
weight_kg	WEIGHT_RANGE	weight_kg <= 0 or > 50 kg → NULL then imputed	19	FLAG	
weight_kg	WEIGHT_IMPUTE	Missing/invalid weight → category median	315	IMPUTE	

list_price	PRICE_FROM_SALES	Invalid list_price re-derived from median deflated selling price	9	REPAIR	
cost_price	COST_OUTLIER	cost_price > 1.2 x list_price (implausible; 1.5x pattern) → re-imputed	67	REPAIR	
cost_price	COST_BLANK	Blank cost_price → imputed	170	IMPUTE	
product_id	DUP_ENTITY	Same product under different IDs (name+category+cost/weight+price within 5%) merged to lowest ID	25	MERGE	3,200 → 3,175

Checked, nothing found: DUP_EXACT, CATEGORY_FROM_SUBCAT, CATEGORY_SUBCAT_CONFLICT, FK_SELLER, PRICE_FROM_COST

customers

Column	Rule	What was wrong → what we did	Rows	Action	Before → after
email	EMAIL_BLANK	Blank email retained as NULL	1,698	RETAIN	
phone	PHONE_NORMALISE	Phone normalised to 03XXXXXXXXX (+92 / dashes / spaces removed)	26,871	STANDARDISE	
gender	STD	Standardised 9 raw variants to 3 canonical values	14,984	STANDARDISE	
gender	STD_BLANK	Blank → 'Unknown'	2,199	IMPUTE	
registration_date	DATE_INVALID	Unparseable registration_date → repaired from first order later	319	FLAG	
birth_year	BIRTH_RANGE	birth_year outside 1940-2010 → NULL	142	FLAG	
birth_year	BIRTH_BLANK	Blank birth_year retained as NULL (age band 'Unknown')	4,600	RETAIN	
city_id	CITY_BLANK	Blank city_id → Unknown location	2,233	IMPUTE	
customer_id	DUP_ENTITY	Same customer under different IDs (name + email or phone, no conflicting birth year/city) merged to lowest ID	2,500	MERGE	45,000 → 42,500
registration_date	REG_AFTER_FIRST_ORDER	Registration after first order → set to first order date	9	REPAIR	
registration_date	REG_FROM_FIRST_ORDER	Unparseable registration date → first order date	226	REPAIR	

Checked, nothing found: DUP_EXACT, EMAIL_FORMAT, PHONE_INVALID, FK_LOCATION

orders

Column	Rule	What was wrong → what we did	Rows	Action	Before → after
*	DUP_EXACT	Exact duplicate rows removed (first kept)	3,200	QUARANTINE	163,200 → 160,000
order_status	STD	Standardised 20 raw variants to 4 canonical values	39,653	STANDARDISE	
channel	STD	Standardised 12 raw variants to 3 canonical values	23,830	STANDARDISE	
discount_code	DISCOUNT_CODE_STD	Codes trimmed/upper-cased (save15 → SAVE15); pct parsed from code	6,062	STANDARDISE	
order_date	DATE_INVALID	Impossible order_date (e.g. 2024-02-30, month 13, 00/00/2024)	1,124	FLAG	
order_date	DATE_OUT_OF_RANGE	order_date outside 2024-01-01..2025-06-30 (2026/2027 typos)	481	FLAG	
order_date	DATE_REPAIRED	Repaired from earliest payment/shipment date of the order	1,605	REPAIR	
customer_id	CUSTOMER_REMAP	Orders re-pointed from duplicate customer IDs to master ID	3,582	MERGE	
customer_id	FK_CUSTOMER	customer_id not in customers → Unknown customer member	321	FLAG	
ship_city_id	SHIP_CITY_FROM_CUSTOMER	Blank ship_city_id → customer's registered city (89% of orders ship to it); flagged	6,082	IMPUTE	
ship_city_id	FK_LOCATION	Ship city still unknown → Unknown location member	306	FLAG	
order_total	TOTAL_INVALID	order_total zero/sentinel/negative/non-numeric → NULL (only used for reconciliation)	488	FLAG	

Checked, nothing found: DISCOUNT_CODE_INVALID, DATE_UNREPAIRABLE

order_items

Column	Rule	What was wrong → what we did	Rows	Action	Before → after
product_id	PRODUCT_REMAP	Lines re-pointed from duplicate product IDs to master	2,793	MERGE	
product_id	FK_PRODUCT	product_id not in catalogue → Unknown product member	932	FLAG	
quantity	QTY_TEXT	Quantity written as word / '2.0' converted to integer	5,767	STANDARDISE	
quantity	QTY_NEGATIVE	Negative quantity → absolute value (sign error)	307	REPAIR	
quantity	QTY_OUTLIER	Quantity 0, non-integer or > 50 (500 = entry error) → re-derived	641	REPAIR	
unit_price	PRICE_FORMAT	'Rs. 2,290' style strings parsed to numbers	7,998	STANDARDISE	
unit_price	PRICE_X100	unit_price ~100x list price (decimal shift) → divided by 100	298	REPAIR	
unit_price	PRICE_NEGATIVE	Negative unit_price → absolute value	316	REPAIR	
unit_price	PRICE_USD	unit_price ~1/280 of list price (recorded in USD) → re-derived in PKR	307	REPAIR	
unit_price	PRICE_INVALID	unit_price 0 / 9999999 / implausible vs list price → re-derived	636	REPAIR	
quantity	QTY_FROM_TOTAL	Unknown quantity solved from order_total (only unknown in order)	627	REPAIR	
unit_price	PRICE_FROM_TOTAL	Unknown unit price solved from order_total (only unknown in order, within 10% of expected)	927	REPAIR	
quantity	QTY_FALLBACK	Quantity not recoverable from order total → 1 (modal value), flagged	14	IMPUTE	
unit_price	PRICE_FALLBACK	Price not recoverable from order total → list price x inflation index, flagged	16	IMPUTE	
discount_amount	DISC_BLANK	Blank discount_amount → order discount % x line gross	31,567	IMPUTE	
discount_amount	DISC_EXCEEDS_GROSS	Discount larger than line gross (e.g. 130% of gross) → recomputed	871	REPAIR	
discount_amount	DISC_MISMATCH	Discount inconsistent with order discount code → recomputed from code	1	REPAIR	
*	ORDER_TOTAL_RECON	Orders whose cleaned line net still differs from order_total by >1% (flagged)	7	FLAG	

Checked, nothing found: DUP_EXACT, FK_ORDER, FK_SELLER

payments

Column	Rule	What was wrong → what we did	Rows	Action	Before → after
*	DUP_EXACT	Exact duplicate rows removed (first kept)	3,296	QUARANTINE	168,111 → 164,815
payment_method	STD	Standardised 22 raw variants to 5 canonical values	41,267	STANDARDISE	
payment_status	STD	Standardised 17 raw variants to 4 canonical values	41,524	STANDARDISE	
amount	AMOUNT_NEGATIVE	Negative amount → absolute value	663	REPAIR	
amount	AMOUNT_SENTINEL	9999999 sentinel → NULL (amount not used for revenue)	162	FLAG	
payment_date	DATE_INVALID	Impossible payment_date → NULL	1,067	FLAG	
payment_date	DATE_BLANK	Blank payment_date retained as NULL	4,927	RETAIN	

Checked, nothing found: FK_ORDER

shipping

Column	Rule	What was wrong → what we did	Rows	Action	Before → after
courier	STD	Standardised 22 raw variants to 6 canonical values	30,368	STANDARDISE	
delivery_status	STD	Standardised 15 raw variants to 3 canonical values	37,699	STANDARDISE	
shipped_date	DATE_INVALID	Impossible shipped_date → NULL	1,022	FLAG	
shipped_date	DATE_BLANK	Blank shipped_date retained as NULL	5,943	RETAIN	
delivered_date	DATE_INVALID	Impossible delivered_date → NULL	924	FLAG	
delivered_date	DATE_BLANK	Blank delivered_date retained as NULL	18,841	RETAIN	
delivered_date	DELIVERED_BEFORE_SHIPPED	delivered_date earlier than shipped_date → duration NULL	1,269	FLAG	

delivered_date	DELIVERY_OUTLIER	Delivery > 30 days after shipping (90-day pattern = date error) → duration NULL	266	FLAG
delivered_date	DELIVERY_VS_ORDER	delivered_date before order_date or > 30 days after it → duration NULL	1,356	FLAG
shipping_fee	FEE_NEGATIVE	Negative fee → absolute value	159	REPAIR
shipping_fee	FEE_SENTINEL	9999999 / > PKR 20,000 fee → imputed	166	IMPUTE
shipping_fee	FEE_IMPUTE	Missing/invalid fee → median fee for same courier & order-weight band	9,311	IMPUTE

Checked, nothing found: DUP_EXACT, FK_ORDER, SHIP_BEFORE_ORDER

returns

Column	Rule	What was wrong → what we did	Rows	Action	Before → after
order_item_id	ID_PADDING	Short order_item_id (OI94118) re-padded to OI00094118	139	REPAIR	
return_status	STD	Standardised 10 raw variants to 3 canonical values	3,993	STANDARDISE	
return_reason	STD	Standardised 38 raw variants to 10 canonical values	7,229	STANDARDISE	
return_reason	STD_BLANK	Blank → 'Not specified'	2,390	IMPUTE	
return_date	DATE_INVALID	Impossible return_date → NULL (Unknown date key)	136	FLAG	
return_date	RETURN_BEFORE_ORDER	return_date before order_date → NULL	155	FLAG	
refund_amount	REFUND_CAP	Refund > line net value by more than 1% (1.4x pattern) → capped at line net	321	REPAIR	
refund_amount	REFUND_ROUNDING	Refund exceeds line net by rounding only (<=1%) → set to line net	5	REPAIR	
*	RETURN_ON_CANCELLED	Return against a cancelled order line - cannot be valid; excluded from returns	15	QUARANTINE	16,048 → 16,033
return_status	RETURN_VALID	Valid returns (status not Rejected/Declined)	14,711	RETAIN	

Checked, nothing found: DUP_EXACT, FK_ORDER_ITEM, REFUND_NEGATIVE, REFUND_BLANK

6. Reject / quarantine table (from `dq.reject_records`)

Table	Reason code	Detail	Records
customers	MERGED_DUPLICATE	Duplicate customer merged into master_customer_id (kept in customer_id_map)	2,500
orders	EXACT_DUPLICATE	Exact duplicate row; first occurrence kept	3,200
payments	EXACT_DUPLICATE	Exact duplicate row; first occurrence kept	3,296
products	MERGED_DUPLICATE	Duplicate product merged into master_product_id (kept in product_id_map)	25
returns	RETURN_ON_CANCELLED_ORDER	Line was cancelled, so it cannot be returned	15

7. Automated quality checks (task `quality_checks`)

Check	Severity	Observed	Threshold	Result	Meaning
fact_rowcount_matches_staging	CRITICAL	315143	315143	PASS	FactSales rows = cleaned order lines
fact_min_rows	CRITICAL	315143		PASS	Fact is not empty
fact_no_duplicate_lines	CRITICAL	0	0	PASS	Grain = one row per order line
fact_net_reconciles_to_staging	CRITICAL	0	0.01	PASS	SUM(net_amount) fact vs staging
unknown_date_share	CRITICAL	0	0	PASS	Every line has a valid order date
unknown_customer_share	CRITICAL	0.00200544	0.05	PASS	Lines mapped to Unknown customer
unknown_product_share	CRITICAL	0.00295739	0.05	PASS	Lines mapped to Unknown product
unknown_seller_share	CRITICAL	0	0.05	PASS	Lines mapped to Unknown seller
unknown_location_share	CRITICAL	0.00190707	0.05	PASS	Lines mapped to Unknown location
net_equals_gross_minus_discount	CRITICAL	0	0	PASS	Arithmetic integrity
no_negative_or_invalid_amounts	CRITICAL	0	0	PASS	qty>0, price>0, 0<=discount<=gross
refund_not_above_line_net	CRITICAL	0	0	PASS	ReturnedAmount <= NetAmount
no_returns_on_cancelled_lines	CRITICAL	0	0	PASS	Cancelled lines cannot be returned
shipping_allocation_sums_to_order_fee	CRITICAL	0	0	PASS	No double counting of courier fees

order_dates_within_valid_period	CRITICAL	0	0	PASS	2024-01-01..2025-06-30
no_null_order_dates	CRITICAL	0	0	PASS	Staging orders all dated
unknown_members_present	CRITICAL	5	5	PASS	Each dimension has key -1
dim_customer_complete	CRITICAL	42501	42501	PASS	All master customers loaded
dim_product_complete	CRITICAL	3176	3176	PASS	All master products loaded
flagged_line_share	WARNING	0.25714	0.3	PASS	Lines carrying at least one repair flag
reject_rate_locations	CRITICAL	0	0.02	PASS	Quarantined (non-duplicate) rows / raw rows
reject_rate_sellers	CRITICAL	0	0.02	PASS	Quarantined (non-duplicate) rows / raw rows
reject_rate_products	CRITICAL	0	0.02	PASS	Quarantined (non-duplicate) rows / raw rows
reject_rate_customers	CRITICAL	0	0.02	PASS	Quarantined (non-duplicate) rows / raw rows
reject_rate_orders	CRITICAL	0	0.02	PASS	Quarantined (non-duplicate) rows / raw rows
reject_rate_order_items	CRITICAL	0	0.02	PASS	Quarantined (non-duplicate) rows / raw rows
reject_rate_payments	CRITICAL	0	0.02	PASS	Quarantined (non-duplicate) rows / raw rows
reject_rate_shipping	CRITICAL	0	0.02	PASS	Quarantined (non-duplicate) rows / raw rows
reject_rate_returns	CRITICAL	0.000935	0.02	PASS	Quarantined (non-duplicate) rows / raw rows
order_total_reconciliation	CRITICAL	4.4e-05	0.03	PASS	Orders whose cleaned lines differ from order_total by >1%
fact_min_rows_threshold	CRITICAL	315143	1000	PASS	Minimum expected fact rows

8. Known limitations

- One duplicate product pair (ElecPro Power Strip U248, P02216/P03090) is not merged: both records carry corrupted costs and one has no weight, so the rule cannot confirm the match without using the name alone.
- June 2025 return rate (≈4%) is understated because some June orders are still inside their return window.
- For 932 lines whose product is not in the catalogue, cost is estimated at 60% of the selling price (catalogue median).
- Payment records exist for every order, including cancelled COD orders, and 4,815 orders have split payments; the order's payment method is the one with the largest amount.